aucmat: matrix-first ROC analysis for high-dimensional biomarker screening in R

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Abstract

Summary: The area under the receiver operating characteristic curve (AUC) is the standard metric for evaluating continuous biomarkers against binary outcomes. When screening thousands of molecular biomarkers, researchers need matrix-wide AUC estimation with proper inference, multiplicity adjustment, and stability assessment. We present aucmat, an R package that provides a matrix-first architecture: `aucmat(X, y)` computes direction-preserving AUCs with DeLong or stratified-bootstrap confidence intervals for every column of a numeric matrix in a single call. The package supports Benjamini-Hochberg multiplicity adjustment, three hypothesis-testing frameworks for paired comparisons (superiority, non-inferiority, equivalence), an omnibus Wald test for equality of multiple correlated AUCs via the joint DeLong covariance matrix, and bootstrap rank-stability analysis. Cross-validated screening eliminates same-data optimism. A two-stage Hurdle-AUC model handles zero-inflated data (scRNA-seq, microbiome) where standard AUC fails. Nine simulation engines generate correlated biomarkers with controlled properties and built-in calibration diagnostics. The package provides 17 visualization functions, including ranked discrimination plots, volcano plots, forest plots, ROC curves with confidence ribbons, precision-recall curves, and Hurdle-AUC diagnostics. On a typical proteomics screen of 10,000 biomarkers, aucmat is approximately 7-fold faster than equivalent per-column pROC loops. The package is freely available under the MIT license at https://github.com/vanhungtran/aucmat and https://vanhungtran.github.io/aucmat.
Availability: https://github.com/vanhungtran/aucmat (source), https://vanhungtran.github.io/aucmat (documentation). MIT license. R >= 4.0.0. **Contact:** tranhungydhcm@gmail.com

Introduction

Receiver operating characteristic (ROC) analysis is the dominant framework for evaluating continuous biomarkers against binary outcomes (**??**). Modern molecular profiling technologies routinely measure thousands of features: a proteomics panel may contain 5,000 proteins, a transcriptomics experiment 20,000 genes. The core statistical task – identifying which biomarkers discriminate cases from controls – requires computing and testing an AUC for each feature, adjusting for multiplicity, comparing top candidates, and assessing ranking stability.

The R ecosystem provides excellent point solutions. pROC (**?**) is the gold standard for single-biomarker ROC inference but requires explicit looping for matrix data. precrec (**?**) adds precision-recall curves for imbalanced data. colAUC (**?**) computes column-wise AUCs but provides no inference. Yet no single package integrates matrix-wide screening, DeLong inference, multiplicity adjustment, non-inferiority testing, bootstrap stability analysis, cross-validation, and multivariate simulation into one coherent workflow. aucmat fills this gap.

Methods

Direction-preserving AUC

For biomarker X and binary outcome $Y \in \{0,1\}$, the AUC is $\mathrm{AUC} = P(X_i > X_j \mid Y_i = 1, Y_j = 0)$. Most software reports $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$, silently reversing the biomarker. aucmat preserves the observed direction, reporting $\mathrm{AUC}_{\mathrm{raw}}$ (the Mann-Whitney estimate on the original scale), $\mathrm{AUC}_{\mathrm{strength}} = 0.5 + |\mathrm{AUC}_{\mathrm{raw}} - 0.5|$ (for ranking), and `effect_direction` (`higher_in_positive` or `lower_in_positive`).

DeLong variance and joint covariance

The DeLong variance estimator (**?**) uses placement values $V_{10}(X_i) = n_0^{-1} \sum_{j:Y_j=0} [\mathbf{1}(X_j < X_i) + 0.5 \cdot \mathbf{1}(X_j = X_i)]$ for positive subjects and V_{01} for negatives, giving $\widehat{\mathrm{Var}}(\widehat{\mathrm{AUC}}) = S_{10}/n_1 + S_{01}/n_0$. For p biomarkers on common subjects, the $p \times p$ joint covariance matrix supports an omnibus Wald test of $H_0 : \mathrm{AUC}_1 = \cdots = \mathrm{AUC}_p$.

Hurdle-AUC for zero-inflated data

Standard AUC assumes continuous biomarkers. On zero-inflated data (scRNA-seq, where 62% of values may be zero), the normal assumption breaks and standard AUC degrades to near 0.5. The Hurdle-AUC uses a two-stage model: (1) $P(X_j = 0 \mid Y) = \mathrm{logistic}(\beta_0 + \beta_1 Y)$ models the binary expression decision; (2) standard AUC on the non-zero values captures magnitude discrimination. The composite Hurdle-AUC score combines both stages. Simulation confirms recovery from 0.53 to 0.90 on zero-inflated biomarkers.

Multivariate simulation

Under the equal-variance binormal model, $X_k \mid Y \sim \mathcal{N}(\mu_{kY}, \sigma^2)$ with $\delta_k = \sqrt{2} \cdot \Phi^{-1}(\mathrm{AUC}_k)$. The class-conditional distribution $\mathbf{X} \mid Y \sim \mathcal{N}_p(\boldsymbol{\mu}_Y, \mathbf{R})$ generates correlated biomarkers with parametrized structures (exchangeable, AR(1), block, user-supplied). The Gaussian copula simulator separates correlation structure (Phase 1: rank-transform copula) from discriminative signal (Phase 2: iterative mean perturbation), achieving both accurate correlations and AUCs without the binormal tradeoff.

Implementation

aucmat is implemented in pure R with dependencies on pROC for ROC curve objects, mvtnorm for multivariate normal generation, ggplot2 for visualization, and rlang for tidy evaluation. The package follows a matrix-first design: the primary function `aucmat()` performs six steps in sequence: input validation, outcome normalization, matrix-wide AUC computation, inference, multiplicity adjustment, and result assembly. All primary functions return S3 objects with `print()`, `summary()`, `plot()`, and `as.data.frame()` methods.

Structured error classes (`aucmat_invalid_outcome`, `aucmat_infeasible_targets`, `aucmat_resampling_failure`) allow programmatic error handling. An RNG-safety mechanism (`with_seed()`) saves and restores the global `.Random.seed` on exit, preventing seed leakage across function calls.

Usage

Basic screening

```r
library(aucmat)

# Generate correlated biomarkers
sim <- simulate_auc_matrix(
  n = 500, prevalence = 0.3,
  target_aucs = c(0.9, 0.8, 0.7, 0.65),
  correlation = 0.3, structure = "exchangeable"
)

# Screen all biomarkers in one call
fit <- aucmat(sim$data[, 1:4], sim$data$truth,
  ci = "delong", adjust = "BH")
print(fit)
summary(fit)
```

Paired comparisons with hypothesis testing

```r
# Two-sided superiority (default)
compare_auc(fit, X, y, reference = "X1")

# Non-inferiority: H0: AUC_a is worse than AUC_b by >= 0.05
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "noninferiority", margin = 0.05)

# Equivalence (TOST): are two AUCs within +/- 0.15?
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "equivalence", margin = 0.15)

# Omnibus test: H0: all AUCs equal
compare_auc_global(fit, X, y)
```

Hurdle-AUC for zero-inflated data

```r
# scRNA-seq: 55% zeros in controls, 25% in cases
sim <- simulate_hurdle_auc(
  n = 400, prevalence = 0.3,
  target_hurdle_aucs = c(0.85, 0.72),
  zero_rate_neg = c(0.55, 0.80),
  zero_rate_pos = c(0.25, 0.70)
)
fit_hur <- hurdle_auc(as.matrix(sim$data[, 1:2]), sim$data$truth)
# Standard AUC ~0.53 → Hurdle AUC ~0.90 (+69%)
```

Cross-validation, panels, and power

```r
# Cross-validated screening (honest out-of-fold AUCs)
cv <- cv_aucmat(X, y, n_folds = 5, seed = 42)

# Build a panel from top biomarkers
panel <- fit_auc_panel(X[, top4], y, method = "ridge", n_folds = 5)

# Sample size for a follow-up study
power_auc_matrix(target_aucs = 0.70, power = 0.80, prevalence = 0.3)
```

Benchmarking

Screening throughput. On a desktop workstation (AMD Ryzen 3.5 GHz, 32 GB RAM), aucmat screens 100, 1,000, and 10,000 biomarkers (n=500) in 0.54, 5.02, and 59.76 seconds respectively. The equivalent per-column pROC loop requires 0.68, 6.42, and 65.1 seconds – a 5–8× speedup.

Simulation calibration. Across 27 simulation conditions (n = 100, 300, 500; prevalence = 0.2, 0.3, 0.5; target AUC = 0.65, 0.75, 0.85; 200 replicates each), DeLong 95% confidence interval coverage ranged from 0.898 (at n=100, AUC=0.65) to 0.957 (at n=500, AUC=0.65). Coverage converged to the nominal 0.95 as sample size increased. Power exceeded 0.85 for AUC >= 0.65 at n=100 and reached 1.0 for AUC >= 0.75 at n >= 300.

```
##
## **Coverage of DeLong 95% CI by sample size and AUC:**
```

Table 1: Empirical coverage of DeLong 95% confidence intervals.

	AUC	n=100	n=300	n=500
1	0.65	0.898	0.937	0.957
4	0.75	0.915	0.943	0.933
7	0.85	0.883	0.923	0.905

Comparison with existing packages

Table 2: Feature comparison of R packages for biomarker ROC analysis.

Feature	aucmat	pROC	precrec	colAUC	dtComb
Matrix-first screening	+	-	-	+	-
Direction-preserving AUC	+	-	-	-	-
DeLong inference	+	+	-	-	-
Stratified bootstrap	+	+	-	-	+
Multiplicity adjustment	+	-	-	-	-
One-sided tests	+	+	-	-	-
Paired comparisons	+	+	-	-	+
Non-inferiority/equivalence	+	-	-	-	-
Omnibus Wald test (3+ AUCs)	+	-	-	-	-
Rank stability	+	-	-	-	-
Partial AUC	+	+	-	-	-
Multivariate simulation	+	-	-	-	-
Simulation validation	+	-	-	-	-
Cross-validated AUC	+	-	-	-	+
Panel scores	+	-	-	-	+
Hurdle-AUC (zero-inflated)	+	-	-	-	-
Precision-Recall	+	-	+	-	-
Power analysis	+	+	-	-	-
Multiclass AUC	+	+	+	+	-
ROC smoothing	+	+	-	-	-
Group-stratified screening	+	-	-	-	-

aucmat uniquely provides: (i) matrix-first screening with direction-preserving AUC and multiplicity adjustment; (ii) non-inferiority and equivalence tests for paired AUC comparisons; (iii) an omnibus Wald test for three or more correlated AUCs; (iv) bootstrap rank-stability analysis; (v) a two-stage Hurdle-AUC for zero-inflated data; and (vi) built-in multivariate simulation with calibration diagnostics.

Conclusion

aucmat provides a unified, statistically principled framework for biomarker screening against binary outcomes in R. Its matrix-first design reduces the code burden for high-dimensional ROC analysis from dozens of scripting steps to a handful of function calls. The package is actively developed with planned extensions for time-dependent (survival) AUCs, sparse-matrix backends, and deep-learning panel construction.

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